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Characterization of distal biceps tendon and triceps tendon injuries in National Football League players from 2009 to 2022

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ABSTRACT

Purpose: The purpose of this study is to determine the incidence and impact of distal biceps and triceps tendon injuries in the National Football League (NFL) from the 2009-10 to the 2022-23 seasons. This study explores the impact that player and injury characteristics have on injury risk, return-to-play, and player performance.

Methods: Data from the 2009-10 to the 2022-23 NFL seasons were analyzed for players with distal biceps or triceps tendon tears. Return-to-play (RTP) and performance metrics were recorded for each player during the season before and first two post-injury seasons. Data were analyzed to determine statistically significant differences in proportions using chi-square, Fisher's exact, or McNemar tests. Statistical significance was set at $p < 0.05$.

Results: Fifty tendon ruptures (26 biceps, 24 triceps) were identified. Biceps injuries were more common in defensive players (73.1 %), while triceps injuries predominantly affected offensive players (58.3 %). Significant risk factors for biceps injuries included BMI ≥31 ($p = 0.0008$) and ≥4 seasons of experience ($p = 0.031$, OR = 2.7). Triceps injuries were associated with BMI ≥31 ($p = 0.01$), age ≥26 ($p < 0.0001$), and ≥4 seasons of experience ($p = 0.006$). RTP rates were 73.1 % for biceps and 70.8 % for triceps injuries. However, only 52.6 % and 41.2 % of players with biceps and triceps injuries, respectively, returned to pre-injury performance levels. Younger players (<26 years) and those with fewer years of experience (≤4 years) were more likely to achieve prior performance levels.

Conclusion: Distal biceps and triceps tendon injuries are typically season-ending for NFL players. Significant risk factors include BMI ≥31, ≥4 years of NFL experience, and game exposure, with age ≥26 being an additional risk factor for triceps tendon injuries. RTP rates exceed 70 %, indicating a strong potential for athletes to return to the NFL post-injury.

1. Introduction

Upper extremity injuries are common among National Football League (NFL) players. Upper extremity injuries make up 18 % of all injuries in an NFL season, including injuries to the biceps and triceps tendons.[1] Injuries to the biceps and triceps tendon in NFL players often result in complete rupture, requiring surgery. Previous literature has reported that NFL players experience a reduction in the number games played per year and length of career following biceps and triceps tendon ruptures.[2,3] Studies have shown that there is not a decrease in performance level following return-to-play (RTP) from operative management of a biceps or triceps rupture, but these injuries are often season ending and require significant rehabilitation and time away from sport-related activities.[3–5] These tendons are injured while the biceps or triceps is contracted and an eccentric load is applied to the muscle, typically seen in offensive and defensive lineman.[5,6]

Biceps tendon ruptures can be managed both operatively and non-operatively. Non-operative management results in, on average, a 30 % reduction in flexion strength, 40 % reduction in supination strength, 80 % reduction in supination endurance, and 15 % reduction in grip strength.[7–9] Due to the reduction in performance and function, NFL players often opt for operative management. Like the biceps tendon, the

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triceps tendon can be managed operatively and non-operatively. Most operatively managed triceps tendon ruptures are complete tendon ruptures, as functional and performance outcomes in partial triceps tears are similar to non-operatively managed ruptures in NFL players.[10] NFL players undergoing operative management of biceps and triceps tendon tears played in a fewer number of games and experienced a shortened career length compared to non-injured NFL players.[10] These findings suggest the triceps and biceps tendon ruptures have a significant impact in NFL players.

Despite the impact of biceps and triceps tendon ruptures in NFL players, there is a lack of literature describing the incidence, risk factors, and outcome on player performance. The current study aims to characterize the incidence and impact of distal biceps and triceps tendon ruptures in NFL players over 14 seasons, with secondary aims to identify how player demographics and injury characteristics influence RTP and performance.

2. Methods

2.1. Data collection and injury identification

NFL player injury data was gathered from publicly available sources.[11–13] NFL injury reports were screened for any players who were listed as "Out", "Injured Reserve", or "Physically Unable to Perform" for at least one game between the 2009–2010 to 2022–2023 NFL seasons.[13] Players who were designated as one of these aforementioned tags did not play in the following game.

All players who missed at least one game were further screened for a biceps tendon or triceps tendon injury. At least two news articles or reports were required to include players in the current study. Data was collected on player biodemographic data (age, height, weight, BMI) and player specific data (experience, position, offense vs. defense) were identified by publicly available sources. This form of data collection is well established in the existing literature.[14–19]

2.2. Player performance

Player performance data was collected from Pro Football Focus (PFF).[20] PFF grades are given to players for every play. PFF grades are then finalized by "senior analysts" who make up the top 2 %–3 % of all analysts. Grades are then accumulated through the duration of the season and converted to a 0–100 scale, allowing for comparison between players.

2.3. Return-to-Play

RTP was defined as a player returning to play in at least one regular season game for at least one in-game play after an injury. RTP data was determined by looking at PFF player performance data and season statistics on Pro-Football-Reference.com.[13] The date of RTP was recorded and the week in the NFL season was recorded. The number of games each player started and played in were recorded for the season prior, after, and two season after injury. The relative number of games played and started were calculated for one and two seasons after injury by dividing post-injury data by data from the season immediately prior to injury. Players who returned to play in as many games and started in as many games within one or two seasons after injury were recorded. RTP time was recorded for each player by determining the difference in dates between injury date and next regular season NFL game played in date.

2.4. Statistical analysis

Players were categorized into position groups, including, offensive line positions (tackle, guard, and center), offensive skill positions (quarterback, running back, wide receiver, and tight end), defensive line positions (defensive end, defensive tackle, and nose tackle), defensive skill positions (cornerbacks, safeties, and linebackers), and special teams. Cutoffs for age (26 years), playing experience (4 years), and BMI (31 kg/m^2), were determined based on NFL averages.[21–24] These cutoffs have been used to determine risk factors for injury in prior NFL studies.[25] Descriptive statistics were calculated to determine means, standard deviations, and percentages. Data were analyzed to determine statistically significant differences in proportions using chi-square, Fisher's exact, or McNemar tests. Statistical significance was set at $p < 0.05$. Odds ratios were calculated when significant differences in proportions were noted.

3. Results

A total of 26 biceps tendon tears and 24 triceps tendon tears were identified in 50 NFL players between the 2009–2010 and 2022–2023 NFL seasons. NFL players with biceps tendon tears had a mean age of 27.46 ± 2.92, mean BMI of 34.81 ± 4.07, and mean years of NFL experience of 5.23 ± 2.72. NFL players with triceps tendon tears had a mean age of 28.74 ± 3.27, mean BMI of 34.65 ± 4.02, and mean years of NFL experience of 7.41 ± 3.40. Patient demographics are listed in Table 1. A graph of incidence of biceps and triceps tendon tears is displayed in Fig. 1.

Of the 26 biceps tendon tears, 22 (84.6 %) occurred in a game and 4 (15.4 %) occurred during practice. Out of 26 biceps tendon tears, 18 (69.2 %) occurred during the regular season and 13 (72.2 %) occurred before week 9. A majority, 20 (76.9 %), of biceps tendon tears were operatively managed.

Of the 24 triceps tendon tears, 20 (83.3 %) occurred in a game and 4 (16.7 %) occurred in a practice. Out of 24 triceps tears, 19 (79.2 %) occurred during the regular season and 13 (68.4 %) occurred before week 9. Like biceps tendon tears, a majority, 17 (70.8 %), of triceps tendon tears were operatively managed. Injury setting, timing, and management is summarized in Table 2.

Seven (26.9 %) biceps tendon tears occurred in offensive players and 19 (73.1 %) occurred in defensive players. Among 7 biceps tendon tears that occurred in offensive players, all occurred among offensive linemen, and none occurred among offensive skill position players. Among the 19 biceps tendon tears that occurred in defensive players, 11 (57.9 %) occurred among players on the defensive line and 8 (42.1 %) occurred among defensive skill position players.

Offensive players suffered 14 (58.3 %) triceps tendon tears and 10 (41.7 %) defensive players suffered biceps tendon tears. Among the offensive triceps tendon tears, 11 (45.7 %) occurred in offensive linemen and 3 (12.5 %) occurred in offensive skill positions. Among the defensive triceps tendon tears, 7 (29.2 %) occurred in offensive linemen and 3 (12.5 %) occurred in offensive skill positions.

No biceps tendon tears or triceps tendon tears occurred among special teams players. Injuries by position and position groups are

Table 1
Patient demographics for biceps tendon tears, triceps tendon tears, and combined. The mean player age was calculated based on player age at the time of injury. The mean years of NFL experience were calculated by determining the total number of seasons that each player participated in at least one game at the time of injury.

	Mean Age	Mean Height (inches)	Mean Weight (lbs)	Mean BMI	Mean Years of NFL Experience
Biceps Tendon	27.46	74.30	274.40	34.81	5.23
std dev	*2.92*	*2.53*	*39.95*	*4.07*	*2.72*
Triceps Tendon	30.12	75.33	281.45	34.65	7.41
std dev	*2.99*	*2.91*	*45.66*	*4.02*	*3.40*
Combined	28.74	74.80	277.80	34.73	6.28
std dev	*3.27*	*2.79*	*42.93*	*4.05*	*3.25*

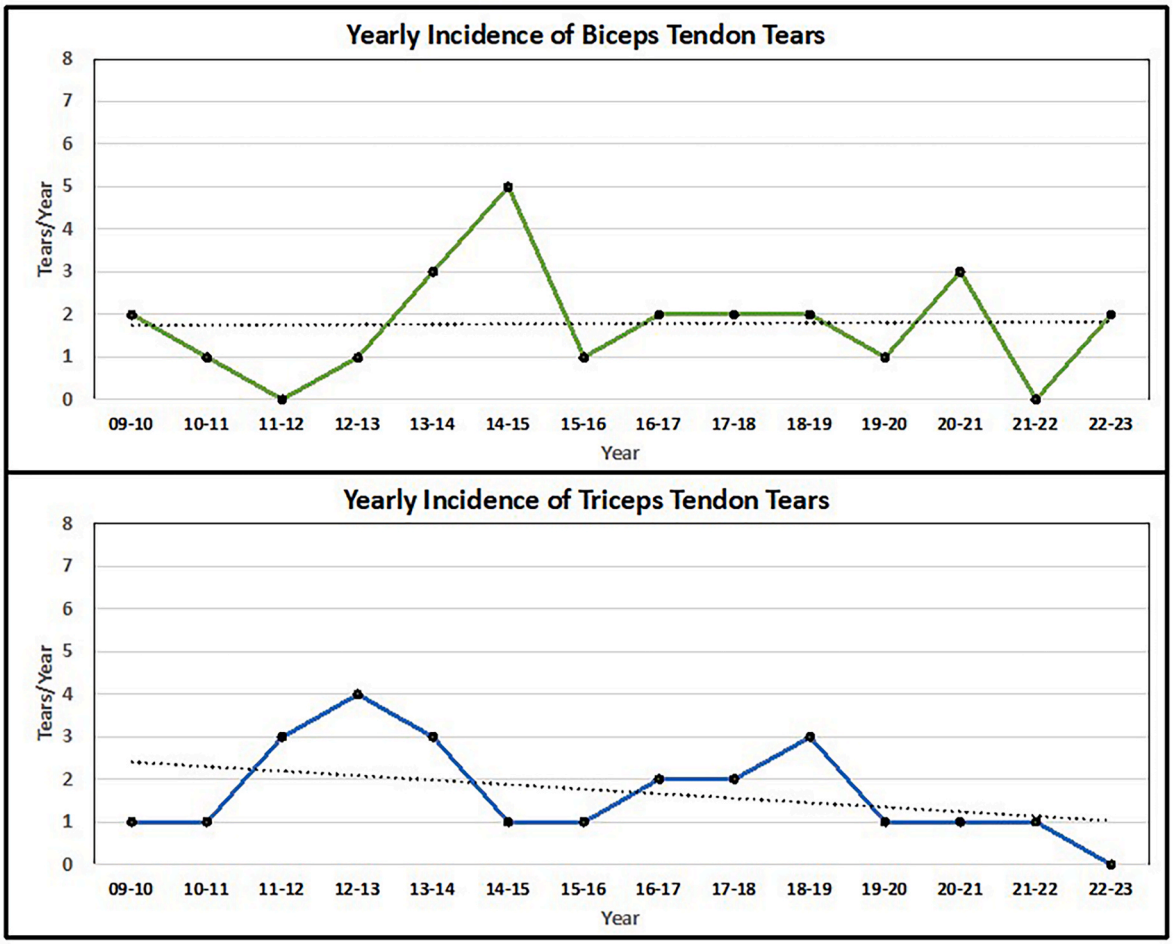

Fig. 1. Yearly incidence for biceps tendon and triceps tears from the 2009–2010 to the 2022–2023 NFL seasons. The dotted black line in each figure represents a linear trendline.

Table 2
Injury characteristics (setting, in-season timing, and management) are displayed for biceps tears, triceps tears, and combined.

	Total Injuries	Game vs Practice		Time of Season that Injury Occurred				
				Preseason Injury	Regular Season Injury			Playoff Injury
		Game	Practice		Game 1-8	Game 9-17	Total	
Biceps Tendon	26	22	4	7	13	5	18	1
%	*52 %*	*84.6 %*	*15.4 %*	*26.9 %*	*72.2 %*	*27.8 %*	*69.2 %*	*3.9 %*
Triceps Tendon	24	20	4	5	13	6	19	0
%	*48 %*	*83.3 %*	*1.7 %*	*20.8 %*	*68.4 %*	*31.6 %*	*79.2 %*	*0.0 %*
Combined	50	42	8	12	26	11	37	1
%	*100 %*	*84.0 %*	*16.0 %*	*24.0 %*	*70.3 %*	*29.7 %*	*74.0 %*	*2.0 %*

summarized in Table 3.

Biceps tendon tears were significantly more likely to occur in NFL players with a BMI ≥31 ($p = 0.0008$, OR = 5.5), more than 4 seasons of experience ($p = 0.031$, OR = 2.7), and during NFL games ($p = 0.0025$, OR = 12.8). Triceps tendon tears were significantly more likely to occur in NFL players with a BMI ≥31 ($p = 0.01$, OR = 5.5), age ≥26 ($p < 0.0001$, OR = 11), more than 4 seasons of experience ($p = 0.006$, OR = 4.5), and during NFL games ($p = 0.015$, OR = 5.6). The age of players was not found to be significant for biceps tendon tears. Position grouping, in-season timing, and pre-, regular-, or post-season timing were not found to be significant for either biceps or triceps tendon tears. Risk factor data is summarized in Table 4.

Between biceps and triceps tendon tears, biceps tendon had a higher RTP rate of 73.1 % (19 out of 26) compared to triceps tendon tears with 70.8 % (17 out of 24). Of the 19 biceps tendon tears, 10 (52.6 %) returned to a prior level of performance, 6 returned to a prior level 1 year following injury and 4 returned to a prior level 2 years following injury. Nine (47.4 %) players did not return to a prior level of performance, and 62.5 % (10 out of 16) returned to play in as many games and started in as many games compared to pre-injury. Among the 17 triceps tendon tears, 7 (41.2 %) returned to a prior level of performance, 5 returned to a prior level 1 year following injury and 2 returned to a prior level 2 years following injury. Ten (58.8 %) players did not return to a prior level of performance, 12.5 % (3 out of 24) returned to start in as many games, and 29.2 % (7 out of 24) returned to play in as many games compared to pre-injury. Average number of days to RTP differed by about 11 days. The average RTP for biceps tendon tears was 332.26 ± 76.83 while the average RTP for triceps tendon tears was 343.62 ±

Table 3
Biceps and triceps tendon tears by position are summarized. Player position groups were broken into offensive linemen, offensive skill positions (quarterbacks, running backs, tight ends, wide receivers), defensive linemen, defensive skill positions (cornerbacks, safeties, linebackers), and special teams.

	Injuries By Position								
	Quarterback	Running Back	Tight Ends	Wide Receiver	Offensive Line	Defensive Line	Defensive Back	Line Backer	Kicker
Biceps Tendon	0	0	0	0	7	11	4	4	0
%	*0.0 %*	*0.0 %*	*0.0 %*	*0.0 %*	*26.9 %*	*42.3 %*	*15.4 %*	*15.4 %*	*0.0 %*
Triceps Tendon	**1**	**1**	**0**	**1**	**11**	**7**	**2**	**1**	**0**
%	*4.2 %*	*4.2 %*	*0.0 %*	*4.2 %*	*45.7 %*	*29.2 %*	*8.3 %*	*4.2 %*	*0.0 %*
Combined	**1**	**1**	**0**	**1**	**18**	**18**	**6**	**5**	**0**
%	*2.0 %*	*2.0 %*	*0.0 %*	*2.0 %*	*36.0 %*	*36.0 %*	*12.0 %*	*10.0 %*	*0.0 %*
	Offensive Skill Positions				**Offensive Line**	**Defensive Line**	**Defensive Skill Positions**		**Special Teams**
Biceps Tendon	**0**				**7**	**11**	**8**		**0**
%	*0.0 %*				*26.9 %*	*42.3 %*	*30.8 %*		*0.0 %*
Triceps Tendon	**3**				**11**	**7**	**3**		**0**
%	*12.5 %*				*45.7 %*	*29.2 %*	*12.5 %*		*0.0 %*
Combined	**3**				**18**	**18**	**11**		**0**
%	*6.0 %*				*36.0 %*	*36.0 %*	*22.0 %*		*0.0 %*
	Offense					**Defense**			**Special Teams**
Biceps Tendon	**7**					**19**			**0**
%	*26.9 %*					*73.1 %*			*0.0 %*
Triceps Tendon	**14**					**10**			**0**
%	*58.3 %*					*41.7 %*			*0.0 %*
Combined	**21**					**29**			**0**
%	*42.0 %*					*58.0 %*			*0.0 %*

184.17. RTP and performance data are summarized in Table 5.

Among players with biceps tendon tears, players aged <26 ($p = 0.0002$, OR = 4.7) and players with ≤4 years of NFL experience ($p = 0.012$, OR = 19) are significantly more likely to return to prior levels of performance. Among players with triceps tendon tears, players aged ≥26 ($p = 0.005$, OR = 60), ≤4 years of NFL experience ($p = 0.42$, OR = 13), and BMI <31 ($p = 0.034$, OR = 14) are significantly more likely to return to prior levels of performance. Players aged <26 are significantly more likely to return to start compared to players ≥26 ($p = 0.032$, OR = 40). Players with a biceps tendon tear are significantly more likely to return to start compared to players with a triceps tendon tear ($p = 0.05$, OR = 60). Comparison data is summarized for biceps tendon and triceps tendon tears in Tables 6 and 7.

4. Discussion

Biceps and triceps tendon tears can have a profound impact on an athlete's career, despite remaining relatively rare among NFL players. Return-to-play data reflects the consequences of biceps and triceps tendon injuries. From the 2009–2010 to the 2022–2023 NFL seasons, there was a similar RTP percentage for both biceps (73.1 %) and triceps (70.8 %) tendon injuries, although players with biceps tendon injuries had a slightly higher RTP rate. More than half of the players with biceps tendon injuries returned to prior performance, while only 41.2 % of players with triceps tendon injuries returned to prior performance. Players with triceps tendon injuries were significantly less likely to return to start as many games when compared to those with biceps tendon injuries. This may indicate that biceps tendon injuries have a higher rate of recovery for NFL players overall. Of the players with biceps tendon injuries who returned to prior performance, 60 % of them returned 1-year post-injury, and 40 % returned 2 years post-injury. Of the players with triceps tendon injuries who returned, 71.4 % of them returned 1-year post-injury, and 28.6 % returned 2 years post-injury. Player characteristics potentially contribute to their ability to return to play. Players with biceps tendon injuries were significantly more likely to return to prior performance if they were <26 years old or had ≤4 years of NFL experience. Players with triceps tendon injuries were significantly more likely to return to prior performance and return to start as many games if they were ≥26 years old. They were significantly more likely to return to prior performance if they had ≤4 years of NFL experience or had a BMI of ≤31.

The incidence of biceps injuries was more than twice the amount in defensive players when compared to offensive players. The incidence of triceps injuries was ~1.4 times more likely in offensive players than defensive players. Biceps and triceps injuries were more common in linemen than skills players both on offense and defense. Special teams' players had zero incidence of biceps or triceps injury in the study timeline. Neither biceps or triceps injuries were significantly more likely to occur in the regular season when compared to the preseason and postseason, although a large majority of the injuries took place during the regular season. Furthermore, neither injury type was significantly more likely to occur during the early regular season when compared to the late regular season, although most injuries took place during the early regular season.

Results of several studies suggest that NFL players with biceps injuries have similarities. Pagani et al. reported a mean age of 28 ± 3 years and mean NFL career experience of 6.0 ± 2.8 years in players with surgically treated distal biceps ruptures, compared to 27.46 ± 2.92 years old and 5.23 ± 2.72 experience years in the present study.[3] Likewise, McGinniss et al. reported a mean age of 28.5 ± 3.5 years in their study on players with surgically treated distal biceps ruptures, although experience years differed (8.3 ± 3.3 seasons in defensive players and 9.6 ± 2.4 seasons in offensive players).[5] In this study, the incidence of biceps injuries was higher among defensive players than offensive players, with 73.1 % of the injuries occurring in defensive players. This is consistent with a previous study from Pagani et al., where 76 % of distal biceps tears occurred in defensive players.[3] Among players with biceps injuries, linemen accounted for a large majority of the cases (68 %), and offensive linemen accounted for all offensive injuries.[3] This data is consistent with the present study, where 69.2 % of players with biceps injuries were linemen, and offensive linemen accounted for all offensive biceps injuries. McGinniss et al. reported similar findings, as most of their subjects undergoing distal biceps repairs were offensive linemen.[5]

A total of 94 % and 84 % of players surgically treated for distal biceps ruptures were able to return to play in McGinniss et al. and Pagani et al., respectively.[3,5] Only 73.1 % of players with bicep tendon injuries returned to play in our study. In this study, 62.5 % of offensive players and 83.3 % of defensive players returned to play compared to 83.3 % of offensive players and 84.2 % of defensive players in Pagani et al.[3] RTP rates in the current study may differ from other studies due to variation in study timelines and protocol. Data from Pagani et al. spans from 1998

Table 4
Risk factors for biceps and triceps tendon tears. Cutoff for significant difference was set at $p < 0.05$, with significant differences highlighted in yellow.

	Risk Factors for Injury													
	Offensive Player	Defensive Player	BMI <31	BMI ≥31	Age <26	Age ≥26	Exerience <4 Years	Exerience ≥4 Years	Game	Practice	Preseason or Playoffs	Regular Season	Early Regular Season	Late Regular Season
Biceps Tendon	7	19	4	22	8	16	7	19	25	1	8	18	13	5
%	*26.9 %*	*73.1 %*	*15.4 %*	*84.6 %*	*33.3 %*	*66.7 %*	*26.9 %*	*73.1 %*	*96.2 %*	*3.8 %*	*30.7 %*	*69.3 %*	*72.2 %*	*27.8 %*
p value	*0.12*		*0.0008*		*0.08*		*0.031*		*0.0025*		*0.920*		*0.0600*	
Triceps Tendon	14	*10*	5	*19*	2	22	4	*18*	22	2	5	19	9	3
%	*58.3 %*	*41.7 %*	*20.8 %*	*79.2 %*	*8.3 %*	*91.7 %*	*18.2 %*	*81.8 %*	*91.7 %*	*8.3 %*	*20.8 %*	*79.2 %*	*75.0 %*	*25.0 %*
p value	*0.22*		*0.01*		*<0.0001*		*0.006*		*0.0150*		*0.260*		*0.100*	

to 2017, and McGinniss et al. reported data from 2000 to 2016. Although the timeline is slightly shorter, the more recent data in this study may better represent current trends. While data from other studies is limited strictly to operative cases, this study contained both operative and non-operative cases, which could have resulted in differences.[3,5] Average RTP time for players with biceps tendon injuries in McGinniss et al. was 324.7 ± 112 days, compared to 332.26 ± 76.83 days in the current study.[5] Although the RTP rate among offensive players was lower in the current study, much of the data regarding biceps injuries in the NFL was similar across studies, indicating that the data in this study may be generalizable.

Triceps tendon injuries presented in NFL players with a mean age of 30.12 ± 2.99 years in the present study, compared to 29 ± 7 years in Mair et al.[10] This may indicate that triceps tendon injuries are more common in players that are slightly older when compared to biceps tendon injuries. Furthermore, Mair et al. reports that triceps tendon injuries are more common among linemen.[10] This was further supported in a study by Finstein et al., where 86 % of the players with surgically treated triceps tendon ruptures were either offensive (40.5 %) or defensive (43.2 %) linemen.[4] The current study found that 75 % of players with triceps tendon injuries were either offensive (45.7 %) or defensive (29.2 %) linemen. RTP data varied among studies on NFL triceps tendon injuries. Surgically treated triceps ruptures had an average RTP period of 165 days with a wide range of 49–318 days, with all players ultimately returning to the NFL.[4] In a study by Mair et al. that reported RTP data in players with triceps tears, all players with complete tears underwent surgery, and eventually gained back full function.[10] In the present study, 70.8 % of players returned to play with an average time to RTP of 343.62 ± 184.17 days, although only 41.2 % of players returned to prior performance. Despite variation among RTP times across studies, most players with triceps tendon injuries were able to return to playing in the NFL.

This study is not without limitations. It is possible that data from the seasons of interest was either missed during data collection or incorrectly reported by the sources. Careful use of several player databases, NFL injury reports, and reliable news outlets allowed for the collection comprehensive biceps and triceps injury data. Overall, player characteristics, such as age, position, and injury outcome were consistent among previously published studies. The results from this study are not generalizable to players playing in their first NFL season as there is no data to compare pre-injury performance level to post-injury performance.

5. Conclusion

Distal biceps and triceps tendon injuries are typically season-ending for NFL players. Significant risk factors include BMI ≥31, ≥4 years of NFL experience, and game exposure, with age ≥26 being an additional risk factor for triceps tendon injuries. RTP rates exceed 70 %, indicating a strong potential for athletes to return to the NFL post-injury.

CRediT authorship contribution statement

Ashwin R. Garlapaty: Conceptualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Molly Jones:** Conceptualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **John Baumann:** Conceptualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Christian Gunn:** Conceptualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **James L. Cook:** Conceptualization, Formal analysis, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Steven F. DeFroda:** Conceptualization, Formal analysis, Investigation, Supervision, Writing – original draft, Writing – review & editing.

Table 5
RTP and performance results for biceps and triceps tendon tears. RTP was defined as returning to play in at least one regular season game after injury.

	Return To Play?		Return to Prior Performance?				Return to start as many games?		Return to play as many games?			Average Time to RTP (days)
			Yes			No						
	Yes	No	Returned in Post-Injury Year 1	Returned in Post-Injury Year 2	Total		Yes	No	Yes	No		
Biceps Tendon	19	7	6	4	10	9	10	16	10	16	**Biceps Tendon**	332.26
%											*std dev*	*76.83*
Triceps Tendon	17	7	5	2	7	10	3	21	7	17	**Triceps Tendon**	343.62
%											*std dev*	*184.17*
Combined	36	14	11	6	17	19	13	37	17	33	**Combined**	337.46
%											*std dev*	*136.90*

Table 6
RTP and return to performance metrics based on player and injury characteristics are displayed for biceps tendon tears. The cutoff for significant difference was set at $p < 0.05$, with significant differences highlighted in yellow.

	RTP			Return to Play as Many Games			Return to Start as Many Games			Return to Prior Performance			Return to Prior Performance in 1 Year or 2 Years		
	Yes	No	p value	Yes	No	p value	Yes	No	p value	Yes	No	p value	1 Year	2 Years	p value
Age <26	7	1		4	4		4	4		6	2		5	3	
Age 26+	12	5	0.62	6	11	0.67	6	11	0.67	0	17	0.0002	1	1	1
Experience ≤4 yrs	9	1		5	5		4	6		5	4		5	3	
Experience >4 years	11	5	0.35	5	10	0.68	6	9	1	1	15	0.012	1	1	1
Offensive Player	5	3		1	6		2	5		1	7		1	0	
Defensive Player	15	3	0.33	9	9	0.18	8	10	0.66	5	12	0.62	5	4	1
BMI ≤31	17	5		1	3		1	3		2	3		1	1	
BMI >31	3	1	1	9	12	0.63	9	12	0.63	4	16	0.56	5	3	1
Biceps Tendon Injury	20	6		10	15		10	15		7	19		6	4	
Triceps Tendon Injury	17	6	1	7	17	0.55	3	21	0.05	5	19	0.74	5	1	0.59
O Line	4	2		1	6		2	5		1	6		1	0	
O Skill	0	0	1	0	0	1	0	0	1	0	0	1	0	0	1
D Line	8	3		6	5		5	6		3	8		3	2	
D skill	7	1	0.6	3	4	1	3	4	1	3	5	1	2	2	1
O line	4	2		1	6		2	5		1	6		1	0	
D line	8	3	1	6	5	0.15	5	6	0.64	3	8	0.62	3	2	1
O Skill	0	0		0	0		0	0		0	0		0	0	
D Skill	7	1	1	3	4	1	3	4	1	3	5	1	2	2	1

Table 7
RTP and return to performance metrics based on player and injury characteristics are displayed for triceps tendon tears. The cutoff for significant difference was set at $p < 0.05$, with significant differences highlighted in yellow.

	RTP			Return to Play as Many Games			Return to Start as Many Games			Return to Prior Performance			Return to Prior Performance in 1 Year or 2 Years		
	Yes	No	p value	Yes	No	p value	Yes	No	p value	Yes	No	p value	1 Year	2 Years	p value
Age <26	14	7		6	15		1	20		2	19		2	1	
Age 26+	14	7	1	1	2	1	2	1	0.032	3	0	0.005	3	1	1
Experience ≤4 yrs	5	0		2	3		2	3		3	2		3	2	
Experience >4 years	23	14	0.15	5	14	0.61	1	18	0.1	2	17	0.042	2	0	0.52
Offensive Player	10	4		3	11		2	12		4	10		4	1	
Defensive Player	18	10	0.74	4	6	0.39	1	9	1	1	9	0.36	1	1	1
BMI ≤31	5	0		2	3		2	3		3	2		3	2	
BMI >31	23	14	0.15	5	14	0.61	1	18	0.1	2	19	0.034	2	0	0.52
Biceps Tendon Injury	20	6		10	15		10	15		7	19		6	4	
Triceps Tendon Injury	17	6	1	7	17	0.55	3	21	0.05	5	19	0.74	5	1	0.59
O Line	7	4		2	9		1	10		2	9		2	1	
O Skill	*3*	0	0.5	1	2	1	1	2	0.4	2	1	0.18	2	0	1
D Line	4	3		3	3		1	5		3	3		1	2	
D skill	3	0	0.47	1	2	1	1	2	1	1	2	1	1	0	1
O line	7	4		2	9		1	10		2	9		2	1	
D line	4	3	1	3	3	0.28	1	5	1	3	3	0.28	1	2	0.99
O Skill	3	0		1	2		1	2		2	1		2	0	
D Skill	3	0	1	1	2	1	1	2	1	1	2	0.99	1	0	1

Ethical statement

Institutional Review Board approval was not needed to conduct this study.

Guardian/patient's consent

Consent for minors was not needed for this study.

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Declaration of competing interests

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